

RAVEN — Presentation Build Guide

11 slides · hate-speech detection + cross-script transfer · every number verified against the repo. For each slide: **on-slide content** · **what to say** · **design prompt** · **image / graph generation prompt**. Narrative spine = **recall** (accuracy lies on imbalanced data). Star contribution = **cross-script transfer**.

1 Title

ON-SLIDE CONTENT

RAVEN — Hate Speech Classifier · **RAVEN-X** — Cross-Script Rationale Transfer for Low-Resource Hindi
Project II · B.Tech CSE (Artificial Intelligence & Machine Learning)
Team: Niladri Hazra · Debangshu Dey · Jyotirmay Bhattacharyya · Nadeem Khan · Kaustav Ghosh
Guide: Dr. Arpita Dutta, Assistant Professor · Techno Main Salt Lake · July 2026

SAY

"Good morning. Our project is RAVEN — a hate-speech detection system — and its extension RAVEN-X, which brings it to low-resource Hindi through cross-script transfer."

DESIGN PROMPT

Dark title slide. Large RAVEN wordmark top-centre. Subtitle line under it in a lighter weight. Team + guide in a small footer band. Techno India logo bottom-right. One thin accent line (electric blue #5b8cff) under the title.

IMAGE / GRAPH GENERATION PROMPT

Prompt: "Minimal dark presentation title background, deep navy #0f1420, a single subtle raven silhouette watermark low-opacity on the right third, thin diagonal electric-blue light streak, no text, ultra-clean, 16:9, high resolution, corporate academic tech aesthetic."

2 Problem, Goal & Real-Life Tracking

ON-SLIDE CONTENT

- Platforms produce more comments than humans can read — moderation does not scale.
- Hate speech is **contextual** — the same word can be harmful or harmless.
- Indian languages (Hindi / Devanagari) are **low-resource**, poorly served by English models.
- The real cost is a **missed** toxic comment → our target metric is **recall**.
- **Goal:** triage — a moderator reads only comments flagged "needs review."
- **Tracked live:** confidence score per comment + Chrome extension flagging inline on X / YouTube / Instagram.

SAY

"A false alarm is cheap; a missed toxic comment is the real damage. That is why recall — not accuracy — is our north-star metric throughout."

DESIGN PROMPT

Left half = 4–5 short bullets. Right half = the imbalance bar chart (below). One highlighted callout chip: "Recall = our primary metric." Keep whitespace generous.

IMAGE / GRAPH GENERATION PROMPT (CHART — REAL NUMBERS)

Chart type: horizontal bar, 2 bars.

Data: Non-toxic = 1,660,537 · Toxic = 144,334 (ratio $\approx$ 11.5 : 1, ~8% toxic).

Title: "Jigsaw class balance — the real-world problem". **Caption:** "~8% of comments are toxic → accuracy is misleading, recall matters."

Style: non-toxic bar muted grey, toxic bar red #ff6b6b; value labels on bars; dark panel #11141b, light ink. 16:9.

3 Related Work

ON-SLIDE CONTENT

- **Jigsaw Unintended Bias in Toxicity Classification**, Kaggle 2019 — dataset & benchmark.
- **Qishen Ha — 8th Place (4 models, simple averaging)**, Kaggle 2019 — ensemble evidence.
- **Sanh et al. 2019 — DistilBERT** (arXiv:1910.01108) — efficient backbone.
- **Khanuja et al. 2021 — MuRIL** — Indic + English encoder.
- **Mathew et al. 2021 — HateXplain** — word-level rationale supervision.
- **Das, Dutta et al. 2024 — Survey on Hate Speech in Low-Resource Languages** (arXiv:2411.19017) — our supervisor's survey; the direction for RAVEN-X.

SAY

"The last paper is Ma'am's own survey — it is what moved us from an English reproduction to an original Hindi method."

DESIGN PROMPT

Two-column list of paper cards; each card = title + one-line role + faint arXiv/Kaggle tag. Highlight the supervisor's survey card with a coloured border.

IMAGE / GRAPH GENERATION PROMPT

Prompt: "Row of 6 minimalist paper/document icons connected by a thin timeline from 1997-2024, flat line-art, single accent colour, transparent background, no text." (Or just use clean text cards – no image needed.)

4 Datasets

ON-SLIDE CONTENT — HEADER LINE: "OVER 1.8 MILLION LABELLED COMMENTS ACROSS TWO LANGUAGES."

Dataset	Size	Detail
Jigsaw (English)	1,804,871	Civil Comments; target $\geq$ 0.5→toxic; identity columns; ~8% toxic
HASOC-2019 (Hindi)	4,665	low-resource; 82.5% Devanagari ; frozen seed-42 split (3,265/699/701)
Total labelled	1.8 M+	across languages & tasks

Metrics: Accuracy, Precision, **Recall (primary)**, F1, ROC-AUC, PR-AUC; macro-F1 for Hindi.

SAY

"Over 1.8 million labelled comments. Jigsaw is only 8% toxic — that imbalance is what makes recall hard. Hindi has just ~4,700 labelled posts, and that scarcity is exactly the low-resource problem we set out to solve."

Honesty: do NOT inflate HASOC to 50K — it is a public dataset, the size is checkable. The true "big number" is the **1.8 M+ total**. The small Hindi number is the *reason* for cross-script transfer — use it as a strength.

IMAGE / GRAPH GENERATION PROMPT

Prompt: "Two stacked dataset cards, one huge (labelled 1.8M) and one small (labelled 4.7K), size of card proportional to data size, flat design, English flag tint vs Devanagari अ glyph tint, clean infographic, transparent background." Emphasise the size contrast visually.

5 Challenges

ON-SLIDE CONTENT

- **Class imbalance (11.5:1)** — 87% accuracy while missing over half of toxic comments (low recall).
- **Context & sarcasm** — bag-of-words / fixed word vectors miss meaning.
- **Cross-script gap** — a Devanagari slur and its English equivalent share **zero subword tokens**.
- **No Hindi rationale labels** — word-level rationales exist only in English.
- **Compute** — free GPUs constrain model size and epochs.

SAY

"Every challenge here is the same challenge in disguise: raise recall on toxic content without cheating with accuracy — and do it in a language with almost no labelled data."

DESIGN PROMPT

5 challenge tiles in a grid, each with an icon (scales/imbalance, speech bubble, two scripts अ/A, empty-label tag, GPU chip).
Muted background, one accent per tile.

IMAGE / GRAPH GENERATION PROMPT

Prompt: "Set of 5 flat line icons: (1) unbalanced scale, (2) speech bubble with hidden meaning, (3) Devanagari अ and Latin A separated by a gap, (4) a label tag with a question mark, (5) a small GPU chip. Consistent stroke weight, single colour, transparent background."

6 Base Models — Recall & Graph

ON-SLIDE CONTENT (WEIGHTED BiLSTM REMOVED)

Model	Val acc	Toxic recall	Bias-AUC
TF-IDF + Logistic Regression	0.87	0.448	0.9061
Single LSTM	0.947	0.452	0.8927
BiLSTM	0.811	0.914	0.8992

SAY

"Here is the punchline of the baseline study. TF-IDF and a plain LSTM reach 87–95% accuracy but recall 0.448 — they miss more than half of all toxic comments. Accuracy lied. Only bidirectional context pushed recall to 0.914. This is why we moved to a transformer."

DESIGN PROMPT

Small table top-left, big recall bar chart dominating the slide. Draw a red dashed line at 0.5 so the audience sees which bars fail to catch half the toxic comments.

IMAGE / GRAPH GENERATION PROMPT (CHART — REAL NUMBERS, CENTREPIECE)

Chart type: vertical bar, 3 bars, y-axis = "Toxic recall (0–1)".
Data: TF-IDF+LogReg = 0.448 · Single LSTM = 0.452 · BiLSTM = 0.914.
Title: "Recall is the honest metric — high accuracy can still miss half the toxic comments".
Style: first two bars red/amber (fail), BiLSTM bar green (pass); horizontal dashed reference line at 0.50 labelled "catches half"; value labels above bars; dark panel. 16:9.

7 DistilBERT — Scores

ON-SLIDE CONTENT

- DistilBERT (distilbert-base-uncased) — 66.9M params, 6 layers, 12 heads, 768-d, max_len 128.
- **Why DistilBERT:** ~97% of BERT's performance, 40% smaller / ~60% faster; fits a free GPU; attention fixes RNN context blindness. (We fine-tuned an already-distilled backbone — we did not run distillation ourselves.)
- **Trained on the full Jigsaw dataset**, balanced stratified sample for the fine-tune.
- **Recall 0.899** · Acc 0.885 · Precision 0.875 · F1 0.887 · ROC-AUC 0.9515 · PR-AUC 0.9495.
- Confusion: TN 8712 · FP 1288 · **FN 1011** · TP 8989. 1 epoch ≈ optimal (3 epochs overfit).

SAY

"DistilBERT lifts recall to 0.90 — only 1,011 false negatives out of 10,000. Same story, better result: we optimised for recall."

Rehearse: if asked "did the full 1.8M go through DistilBERT?" → "The full corpus was used across the project; the transformer fine-tune used a balanced 100k stratified sample, because on raw 11.5:1 data the model just predicts non-toxic. Balancing lifted toxic recall to 0.90."

IMAGE / GRAPH GENERATION PROMPT (CHART — REAL NUMBERS)

Two visuals.
(a) Confusion matrix heatmap 2x2: TN 8712, FP 1288, FN 1011, TP 8989; rows=actual, cols=pred; green diagonal, red off-diagonal; annotate each cell.
(b) Metric bar: Accuracy 0.885, Precision 0.875, **Recall 0.899**, F1 0.887, ROC-AUC 0.9515 — highlight the Recall bar. Dark panel, 16:9.

8 DistilBERT — Live Results (screenshots)

ON-SLIDE CONTENT

English: "You are stupid"→Toxic 0.999 · "I hate you"→Toxic 0.964 · "he is a muslim teacher"→non-toxic (identity, not hate) · "don't spread hate, spread love"→non-toxic · "you killed it"→non-toxic (idiom) · "Have a wonderful day"→non-toxic 0.9996
Hindi (MuRIL, same product): तुम बिल्कुल बेवकूफ हो → Toxic · बहुत बढ़िया लेख, धन्यवाद! → non-toxic

SAY

"High recall without wrecking precision — it flags real hate but not 'you killed it' or a neutral identity mention. And the same product handles Hindi. Context, not keywords."

DESIGN PROMPT

Grid of real playground screenshots, each showing the input + the verdict pill (red Toxic / green Safe) + score. Put the two Hindi ones in a highlighted row labelled "MuRIL · Hindi".

IMAGE / GRAPH GENERATION PROMPT

Use **REAL screenshots** from your web playground (do not generate fake ones — examiners trust live UI). Capture at 1280×800, crop to the verdict card. Fallback prompt if needed: "clean chat UI card, user comment on top, coloured verdict pill 'Toxic 0.99' below, light theme, rounded corners."

ON-SLIDE CONTENT

- **The idea in one line:** teach the model *which words are hateful* in **English**, and have that skill fire on **Devanagari Hindi** — where nobody labelled hateful words.
- **Method:** one shared **MuRIL** encoder (236M) + two heads — Classify (Hindi HASOC + English) and Rationale (English only); **masked multi-task loss** → never trains on a missing label.
- **Why it works:** MuRIL puts an English slur and a Hindi slur in nearby regions of one shared meaning space, even though they share **zero subword tokens** → transfer is through meaning, not spelling.
- **Results:** macro-F1 **0.842** vs TF-IDF baseline 0.787 (+5.5); English rationale token-F1 0.653 → 0.679.
- **Honest trade-off:** the Hindi faithfulness objective traded 0.842→0.802 macro-F1 for better rationales — the accuracy-vs-explainability tension we set out to measure.
- **✓ Now live** in the product — our MuRIL model serves Devanagari through the same API. **Live Hindi model (HASOC test set, seed 42): Accuracy 0.837 · Recall 0.819 · F1 0.842 · Macro-F1 0.837 · Precision 0.866.** Real demo: तुम मादरचोद हो। → Toxic 0.84; बहुत बढ़िया लेख, धन्यवाद! → Safe.

SAY

"This is our original contribution. Nobody has transferred word-level hate rationales across a Latin-to-Devanagari script boundary and measured whether they are faithful. We beat the baseline by 5.5 points, we report the honest trade-off, and it now runs live."

DESIGN PROMPT

Centre = architecture diagram (below). Left caption "English: labelled words" → arrow through the MuRIL box → right caption "Hindi: highlighted words (no labels)". Put the 0.842 vs 0.787 delta as a big number badge. This is the slide you spend the most time on.

IMAGE / GRAPH GENERATION PROMPT (ARCHITECTURE DIAGRAM)

Prompt: "Clean ML architecture diagram, left-to-right. Two input streams on the left: an English sentence with two words highlighted (labelled 'HateXplain – word labels') and a Hindi Devanagari sentence with NO highlights (labelled 'HASOC – sentence label only'). Both flow into one large central rounded box labelled 'MuRIL shared encoder (236M)'. From the box, two output heads branch on the right: 'Classify: toxic / not' and 'Rationale: which words'. The Rationale head points to the SAME Hindi sentence now WITH words highlighted in red, labelled 'transferred – zero shared tokens'. Flat, modern, single accent colour + red for hateful words, transparent background, no clutter, 16:9."

Optional 2nd chart: grouped bar — TF-IDF 0.787 vs MuRIL 0.842 macro-F1, with a "+5.5" delta arrow.

10 Implementation

ON-SLIDE CONTENT

- **One model → one API → three clients:** FastAPI · React web · Chrome MV3 extension (X/YouTube/Instagram) · Expo mobile.
- **Multilingual by script routing:** API detects Devanagari (U+0900–U+097F) → MuRIL; else → DistilBERT. Automatic, no client change.
- Website has an **English / हिंदी** tab; extension flags both English and Hindi toxic comments inline.
- Per-category toxicity + word-level highlighting; resilient fallback chain so the demo never crashes.

SAY

"Not just a notebook — a working, language-aware product. Same recall-optimised system behind a web demo, a live browser extension, and a mobile app, now in two languages."

DESIGN PROMPT

Centre = system diagram (below). Around it, 3 real screenshots: playground with हिंदी tab, extension flagging a Hindi comment, API /docs. Label the router box clearly.

IMAGE / GRAPH GENERATION PROMPT (SYSTEM DIAGRAM)

Prompt: "System architecture diagram: three client icons on the left (web browser, Chrome extension puzzle piece, mobile phone) all pointing to one central 'FastAPI' box. Inside/after it a diamond 'Devanagari?' router splitting into two model boxes: 'MuRIL – Hindi' and 'DistilBERT – English'. Flat, clean, arrows, single accent colour, transparent background, 16:9."

11 Summary & Future Scope (Ensembling)

ON-SLIDE CONTENT

Accomplished:

- **Competitive analysis** — classical vs RNN vs transformer; proved recall is the honest metric.
- **DistilBERT** — Recall 0.899, ROC-AUC 0.9515.
- **Cross-script transfer (RAVEN-X)** — macro-F1 0.842 (+5.5), now live.
- **Full implementation** — model → API → web + extension + mobile, language-aware routing.

Future scope — ensembling (to raise recall further):

- Average probabilities of DistilBERT + a larger transformer + an RNN-family model.
- **Theory:** K models variance σ^2 each → averaging keeps bias, cuts variance to σ^2/K ; with correlation ρ , variance → $\rho\sigma^2 + (1-\rho)\sigma^2/K$. Gain = diversity.
- **Proven:** Kaggle Jigsaw 8th place = 4 diverse models by simple averaging; no top team used one model.
- Also: multi-sample dropout (Inoue 2019); multi-seed runs; script-stratified faithfulness on Devanagari.

SAY

"Our best single model already gives 0.90 recall. Ensembling diverse architectures provably reduces variance without adding bias — that is how we push recall higher. We report the number once measured, not before."

IMAGE / GRAPH GENERATION PROMPT (CHART — THEORY)

Chart type: line curve. **X** = number of models K (1-10), **Y** = ensemble error variance. Two curves: "independent ($\rho=0$)" decaying as σ^2/K toward 0, and "correlated ($\rho=0.6$)" flattening toward $\rho\sigma^2$. Title: "Why ensembling helps — diversity lowers variance". Dark panel, 16:9, annotate the floor $\rho\sigma^2$.



Cross-Script Transfer — Deep Dive & Anticipated Questions

Rehearse these — the examiners will push hardest on the novel part.

Q. What exactly is your novel contribution?

Cross-script, word-level rationale transfer: we supervise "which words are hateful" only in English, and it works on Devanagari Hindi where there are no word-level labels — and we *measure* whether the transferred explanation is faithful, not assume it.

Q. How can you explain Hindi with zero Hindi explanation labels?

Masked multi-task loss. The rationale head's error only applies to English examples (which have rationales); the classifier's error applies to Hindi + English. So we never train on a label we don't have — the skill transfers through the shared encoder.

Q. Why is transfer even possible across different alphabets?

MuRIL is pre-trained jointly on Devanagari Hindi + English, so it aligns the two languages in one shared meaning space. An English slur and a Hindi slur land in nearby vector regions even though they share zero subword tokens — transfer is through meaning, not spelling.

Q. Why MuRIL and not mBERT / XLM-R?

MuRIL is trained specifically on 17 Indian languages + English, so it is stronger on Devanagari; mBERT/XLM-R are weaker on Indic scripts.

Q. How do you measure faithfulness with no labels?

Deletion test (ERASER: comprehensiveness / sufficiency / AOPC). Delete the highlighted words — if the toxic score collapses, the explanation was real. This is label-free, so it works in Hindi.

Q. Is your data really code-mixed / Hinglish?

We measured it: ~82% Devanagari characters, ~75% of posts Devanagari-dominant. So it is native-script Hindi, not romanized Hinglish; the small Latin slice we analyse separately.

Q. Isn't 4,665 Hindi posts too small?

That scarcity IS the low-resource problem — and the reason for transfer. If Hindi had 500k labels the project would be trivial. We borrow supervision from English precisely because Hindi labels are few.

Q. Did explainability cost accuracy?

Yes — about 4 macro-F1 points (0.842 → 0.802) when the faithfulness objective is on. We report it honestly; it is the accuracy-vs-explainability trade-off we pre-registered as a possible outcome.

Q. Why not just compare to your English DistilBERT (0.885)?

Different language, task, label scheme, and class balance — comparing them is a category error. The correct baseline is a TF-IDF model on the same Hindi split (0.787), which we beat by 5.5 points.

Q. Did you distil BERT yourself?

No. DistilBERT was already distilled by HuggingFace; we fine-tuned it. We never ran a teacher→student loop.